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March 12th, 2026



Proposal Document

Max Showdown Trivia Software

Version 1.1

This document outlines the general scope of your project based on our understanding of the requirements provided and discussions to date.

Please don't print me, it's bad for the environment

Prepared For: Max Allan

Lead ID: 200575

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Table of Contents

PROJECT DESCRIPTION	6
KEY FEATURES	7
SCOPE OF WORK	9
1. USER TYPE.....	9
2. GAME FLOW	12
3. HOST CONTROL INTERFACE (WEB — DESKTOP/LAPTOP)	13
3.1 Game Session Dashboard	13
3.2 Game Controls	14
3.3 Music Round Controls	14
3.4 Mini-Game Launcher	14
3.5 Manual Team Management (New)	14
4. PLAYER MOBILE INTERFACE (MOBILE WEB — RESPONSIVE)	15
4.1 Session Join	15
4.2 Lobby / Waiting Screen.....	15
4.3 Question & Answer Screen.....	15
4.4 Answer Reveal & Feedback.....	16
4.5 Wager Input (Wager Rounds).....	17
4.6 Elimination Round Behaviour	18
4.7 Majority Rules Behaviour.....	19
4.8 3D Mini-Game Interaction	19
4.9 Reconnection	19
5. VENUE DISPLAY SCREEN (WEB — PROJECTED/TV)	20
5.1 Welcome Screen (New).....	20
5.2 Team Registration Grid (New)	20
5.3 Round Header	21
5.4 Round Intro Slide (New).....	22
5.5 Question Display	22
5.6 Countdown Timer	23
5.7 Live Response Counter	24
5.8 Answer Reveal	24

5.9	Scoreboard Display	25
5.10	QR Code Display.....	26
5.11	Mini-Game Display	26
5.12	Break Timer (New)	26
6.	ROUND TYPES & GAME LOGIC.....	27
7.	SCORING ENGINE	28
7.1	Base Scoring.....	28
7.2	Wager Round Scoring	28
7.3	Majority Rules Scoring	28
7.4	Elimination Round Scoring (Updated).....	28
7.5	Leaderboard.....	28
8.	MINI-GAMES	29
8.1	3D Horse Race (Updated from 2D)	Error! Bookmark not defined.
8.2	3D Card Shuffle Game (Replacing Cups and Balls).....	29
9.	ADMIN PANEL & CONTENT MANAGEMENT (WEB).....	29
9.1	Quiz Builder.....	30
9.2	Question Bank.....	30
9.3	MP3 Upload & Management.....	30
9.4	Session Configuration	30
10.	SESSION & LOBBY MANAGEMENT	31
10.1	Session Creation.....	31
10.2	Player Join Flow	31
10.3	Lobby	31
10.4	Reconnection	31
11.	BACKEND GAME ENGINE	32
12.	UI/UX & DESIGN SYSTEM	32
13.	AUDIO & SOUND SYSTEM	33
COST ESTIMATE.....		34
PAYMENT SCHEDULE		34
ESTIMATE NOTES.....		34
RESOURCE UTILIZATION		35
TECHNOLOGY SPECIFICATIONS.....		35

CLIENT RECEIVABLES	36
DELIVERABLES	36
IMPORTANT POINTS.....	37
ASSUMPTIONS	37
SOFTWARE DEVELOPMENT LIFE CYCLE:.....	40
DEVELOPMENT PROCESS.....	44
PROJECT MANAGEMENT TOOL	44
TESTING	44
ENVIRONMENT	44
DEPLOYMENT	44
WARRANTY	44
INTELLECTUAL PROPERTY RIGHTS	44
CMMI	45
WHY DOTSQUARES?.....	46
.....	47

Revision History

Version	Date	Description	Prepared by	Reviewed by
1.0	06-03-2026	Initial Analysis & Estimation	Varun Chaudhary	Mohit Kumar
1.1	11-03-2026	Scope updated based on client feedback (Max)	Varun Chaudhary	Mohit Kumar/Chetan Maini

Note: All updates to the scope in this version (v1.1) are highlighted in **blue** for easy reference.

Purpose of the Document

This document outlines the general scope of your project based on the requirements provided and discussions to date. The main objective at this stage is to ensure that all parties agree upon the general scope, budget and schedule of the project prior to commencing work on a time & materials basis.

Clients are urged to ensure that all details contained within this document are clear and fully meet their requirements before any work commences. Timescales mentioned only cover the scope listed, if these changes, costs and timings will change accordingly.

Once you are happy, we will progress into the next stages. This may include further detailing requirements and functionality, or it may go straight into design/development. This will depend on the nature of the project.

Project Description

Based on the client's Developer Handoff Pack, mockups, prototype, and subsequent clarifications, the objective of this project is to develop a real-time, web-based live trivia gameshow platform called Max Showdown.

The platform will consist of three interconnected interfaces: a Host Control Panel for the quiz master, a Player Mobile Interface for audience participation via smartphones, and a Venue Display Screen for projection at live events. A fourth component, a web-based Admin Panel, will allow the host to manage quiz content, build question banks, and configure game sessions.

The host will control the entire game flow — from starting timers and revealing answers to advancing through rounds and displaying scoreboards. Players will join sessions by scanning a QR code, which will redirect them to a server URL with a session PIN. The venue screen will display questions, answer options, countdown timers, and live scoreboards for the audience.

The platform will support seven distinct round types: Multiple Choice, Wager, Music, Elimination, Majority Rules, Final Multiple Choice, and a Final Wager question. It will be designed to handle a minimum of 300 concurrent players with optimal latency.

Two mini-games (Cups and Balls and Digital Horse Race) are included in the scope as embedded entertainment modules that can be launched during a game show.

Following the discovery call, the mini games have been upgraded from 2D to 3D visuals. The Cups and Balls game has been replaced with a 3D Card Shuffle variant. Additional venue displays features (welcome screen, team registration grid, break timer, persistent QR code, live response panel, and auto-reveal) have been added to the scope. The elimination round mechanics have been refined with incremental scoring and a

confirmed “all-teams-wrong” rule. Answer options are now dynamic (2 to 6 per question). These changes are detailed in the scope of work sections below.

Purpose of the Project

The purpose of Max Showdown is to provide a dedicated, custom-built trivia hosting platform that replaces reliance on third-party software (such as QuizXpress) and gives the host full control over the game format, branding, audience interaction, and live show experience.

The platform will enable the host to run professional-quality trivia events at scale, with real-time audience participation, automated scoring, and a polished visual presentation designed for venue projection.

Once the core platform and round structure are built, the host will only need to update quiz content on a weekly basis. The underlying infrastructure, round formats, and scoring logic will remain consistent, enabling efficient multi-venue operation through a shared question bank.

Key Features

- **Three-Interface Architecture:** Host Control Panel, Player Mobile Interface, and Venue Display Screen connected in real time via WebSockets.
- **Admin Panel & Question Bank:** Web-based content management system for creating quizzes, managing a reusable question bank across multiple venues, and configuring game sessions.
- **Host Game Control:** Full control of game progression via on-screen buttons and keyboard shortcuts (Space, T, P, S).
- **QR Code Join:** Players will scan a QR code at the venue to join the session via a server URL and session PIN.
- **Team-Based Play:** The platform is designed for team play. Solo players will join as a team of one.
- **Seven Round Types:** Multiple Choice, Wager, Music, Elimination, Majority Rules, Final Multiple Choice, and Final Wager.
- **Two Mini-Games:** 3D Horse Race and 3D Card Shuffle Game as embedded entertainment modules. Upgraded from 2D to 3D following the discovery call.
- **Scoring Model:** +10 for correct, -2 for incorrect, with specific scoring rules for Wager and Majority Rules rounds.
- **Elimination Round:** Incremental scoring (10 to 120 pts across 12 questions). Teams are eliminated for incorrect answers. If all remaining teams answer incorrectly on the same question, no elimination occurs — all lose points but proceed.
- **MP3 Audio Playback:** Host will upload and play MP3 files through the venue sound system for Music rounds.

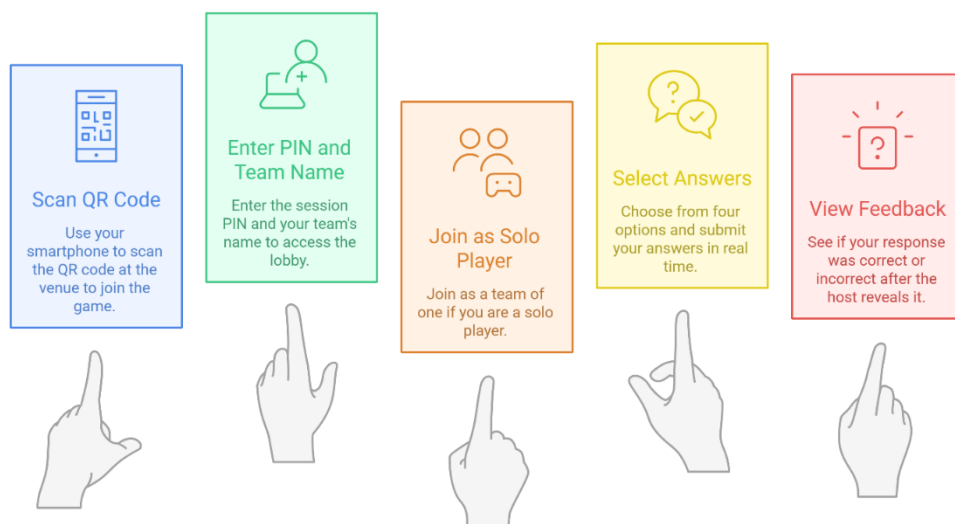
- **MP4 Video Playback:** Host will be able to upload and embed video clips (MP4) into any question slide for music videos, observation questions, or visual content.
- **Venue-Optimised Display**
 - 16:9 widescreen layout with countdown timers, round headers, answer reveal, and leaderboards.
 - Added to the Venue Display Screen. Welcome Screen, Team Registration Grid, Round Intro Slides, Break Timer, Persistent QR Code, Live Response Panel, Auto-Reveal, and Points Display
- **300 Concurrent Players:** Architecture designed for high concurrency with sub-200ms response latency.
- **Reconnection Support:** Disconnected players/teams can rejoin an active session using the session PIN.
- **Dynamic Answer Options:** Questions can have 2 to 6 answer options (not fixed at 4). Supports true/false, standard, and extended question formats.
- **Host Manual Team Management:** The host can manually add, remove, and edit team scores during a live session.
- **Timer Sound:** Timer sound plays throughout all question rounds. Timer sound is muted during Music rounds only to avoid interference with MP3 playback.
- **First Answer is Final:** Once a team submits an answer, it is locked in and cannot be changed.
- **Dark Theme Design:** Apple-like simplicity with dark UI, electric blue accents, smooth animations, and large, clean typography.

Scope of Work

Based on the received requirement, we have elaborated all points as part of this Scope of Work Section:

1. User Type

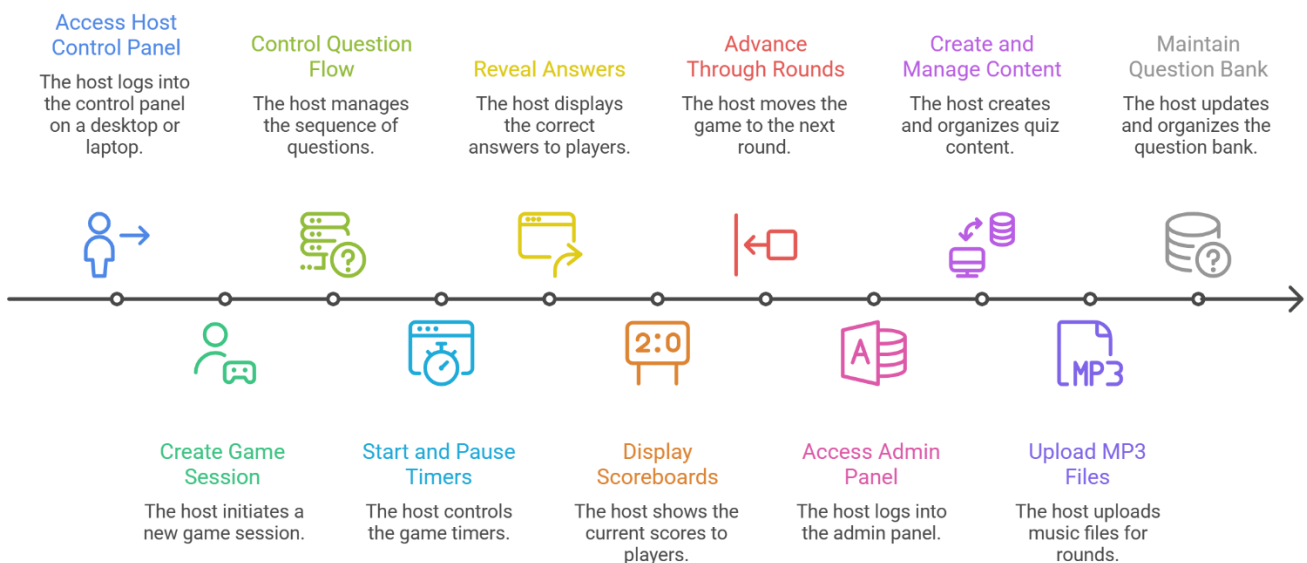
- **Players / Teams (Audience Participants)**
 - Players will use their smartphones to join a live game session by scanning a QR code displayed at the venue.
 - Players will enter a session PIN and their team's name to join the lobby.
 - The platform is designed for team-based play. Solo players will be able to join as a team of one.
 - Each team will use one mobile device as their buzzer. The system only tracks the number of teams, not individual players within a team.
 - Players will view questions, select answers from four options, and submit responses in real time.
 - First answer is final — once a team member taps an option, it is locked in and cannot be changed.
 - No speed bonus — points are not affected by how quickly a team answers.
 - Players will see visual feedback on whether their response was correct or incorrect after the host triggers the reveal.
 - Players will see the countdown timer, question text, and all answer options on their mobile device.



- **Host / Quiz Master:**

- The host will use the Host Control Panel on a desktop or laptop to manage the entire game session.
- The host will create game sessions, control question flow, start and pause timers, reveal answers, display scoreboards, and advance through rounds.
- The host will interact using both on-screen buttons and keyboard shortcuts for seamless live performance.
- The host will use the Admin Panel to create and manage quiz content, upload MP3 files for music rounds, and maintain the question bank.
- The host will be able to manually add a team (with a custom score), remove a team, and edit any team's score during a live session.
- The host will always have a live scoreboard visible on their device before it is displayed on the venue screen.

Host User Flow



- **Venue Audience (Passive Viewers)**

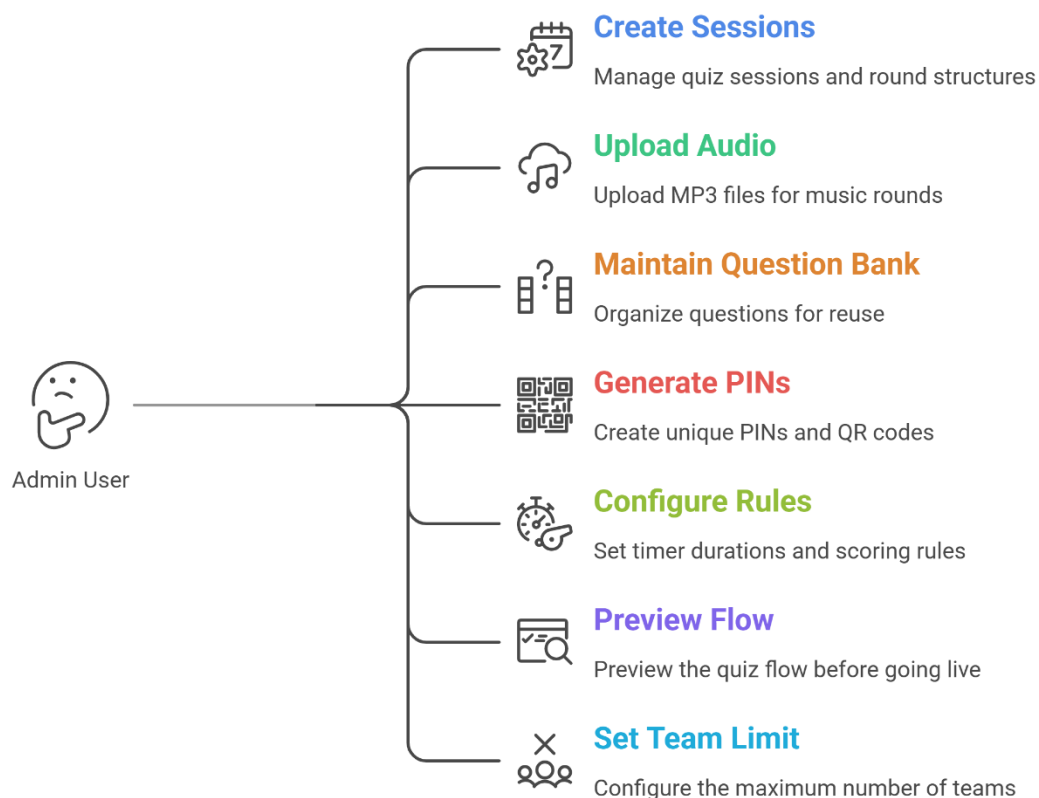
- The venue audience will view the Venue Display Screen projected via a TV or projector at the event location.
- This screen is view-only and controlled entirely by the host.
- All audio (sound effects, music, and MP3 playback) will play through the venue sound system only. No audio will play on player devices.

- **Admin User (Host — Max Allan)**

- The admin will access the web-based Admin Panel to create and manage quiz sessions, round structures, and question content.
- The admin will upload MP3 audio files for use in Music rounds.
- The admin will maintain a central question bank, allowing questions to be categorised, searched, and reused across multiple venues and events.
- The admin will generate unique session PINs and QR codes for each live game session.
- The admin will configure round types, timer durations, and scoring rules per quiz.
- The admin will preview the full quiz flow before going live.
- The admin will be able to set the maximum number of teams per session.

Note: The admin and the host are the same person (Max). The Admin Panel is the back-office tool for preparation; the Host Control Panel is the live performance tool. Both are accessed via a web browser.

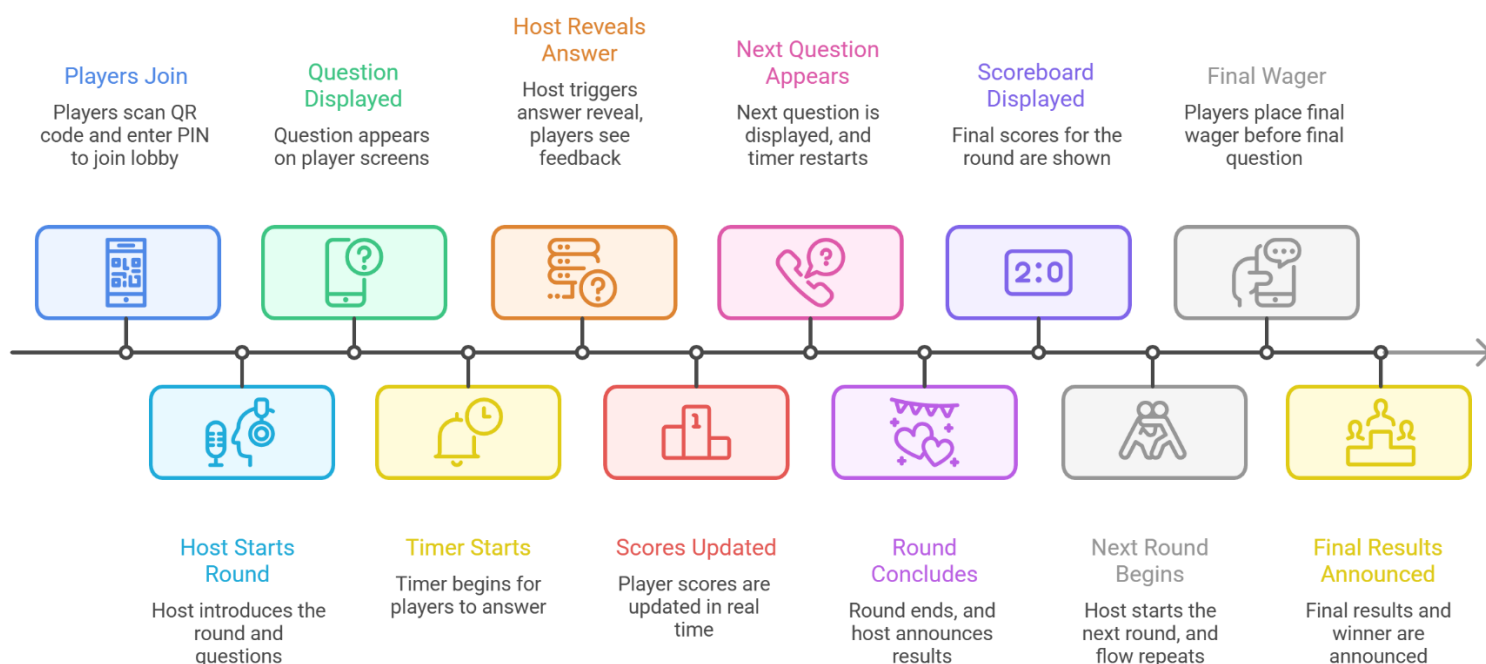
Admin User Flow



2. Game Flow

- The complete game flow, from session creation to final results, will follow the sequence defined in the Developer Handoff Pack:
- Lobby → Round Intro → Question → Timer → Answer Reveal → Score Update → Next Question → Round End → Scoreboard → Next Round → Final Wager → Final Results
- The host will control transitions between each state. No automatic advancement will occur without host action, ensuring the quiz master retains full control of the show's pacing and flow.
- When the host presses Next/Space, three things happen simultaneously: the timer starts, the answer options appear on player phones, and (for music rounds) the MP3 begins playing.
- Auto-Reveal: If all teams submit their answers before the timer expires, the system will automatically reveal the correct answer without waiting for the remaining countdown time.
- All other transitions (scoreboard, next round, mini games) require explicit host action. No slide, question, or scoreboard will auto-advance.

Live Quiz Game Flow

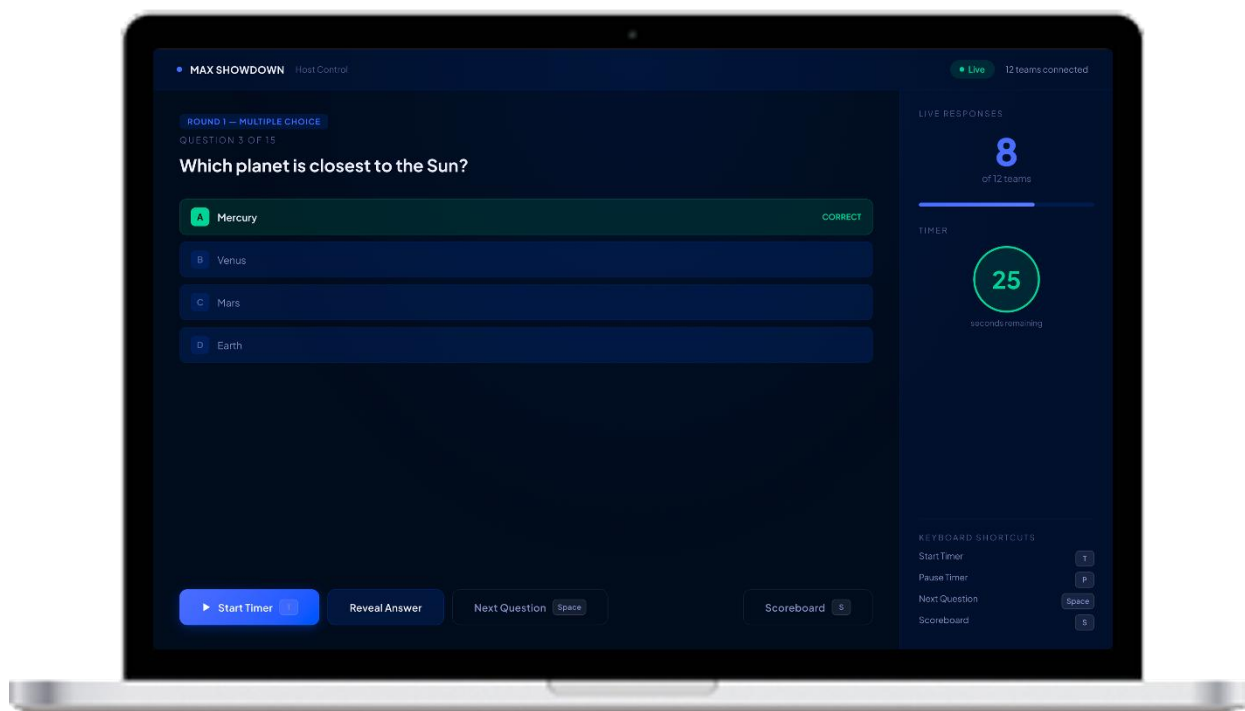


3. Host Control Interface (Web — Desktop/Laptop)

The Host Control Panel will serve as the quiz master's command centre during live events. The interface will be optimised for desktop/laptop usage and designed for speed and simplicity during live performance.

3.1 Game Session Dashboard

- The host will see the current question number and total questions (e.g., "Question 3/15").
- The host will see the full question text displayed prominently.
- The host will see the current round name and type.
- The host will see the number of players/teams currently connected.
- The host will see a live count of responses received for the current question.
- The host will always see a live scoreboard on their device, viewable before it is shown on the venue screen.
- The host will see how many points the current question is worth.



Note: The UI screen mock-ups are shown in this document are for demonstration purposes only. The actual screens during development and deployment may differ.

3.2 Game Controls

- Start Timer: The host will tap the “Start Timer” button or press the T key to begin the countdown.
- Pause Timer: The host will press P to pause the countdown if required.
- Reveal Answer: The host will tap “Reveal Answer” to display the correct answer on the venue screen and trigger score calculation.
- Next Question: The host will tap “Next Question” or press Space to advance to the next question.
- Show Scoreboard: The host will press S to display the current leaderboard on the venue screen.
- **Timer sound:** An audible countdown sound will play through the venue system during all question rounds. Timer sound will be automatically muted during Music rounds to avoid interference with the MP3 audio.

3.3 Music Round Controls

- During Music rounds, the host will see a play/pause control for the MP3 file associated with the current question.
- Audio will play through the host’s laptop output (connected to the venue sound system). No audio will be sent to player devices.
- When the host presses Next, the timer starts, the MP3 plays, and answer options appear on player phones — all simultaneously.
- For slides with embedded video, the host will see play/pause controls. Video playback will display on the venue screen and audio will route through the venue sound system.
- No timer sound during music rounds — the music itself is the audio content.

3.4 Mini-Game Launcher

- The host will be able to launch either mini game (Cups and Balls or Horse Race) at any point during the show via the Host Control Panel.
- The mini game will display on the Venue Display Screen and accept player inputs via mobile devices.

3.5 Manual Team Management (New)

- The host will be able to manually add a new team at any point during the session, including setting a custom starting score.
- The host will be able to manually remove a team from the session.
- The host will be able to manually edit any team’s score (increase or decrease) at any time.
- Use case: If a team disconnects and rejoins, the host can re-add them with their previous score.

4. Player Mobile Interface (Mobile Web — Responsive)

The Player Mobile Interface will be a responsive web application accessed via smartphones. Players will not need to download a native app — they will access the game through their mobile browser.

4.1 Session Join

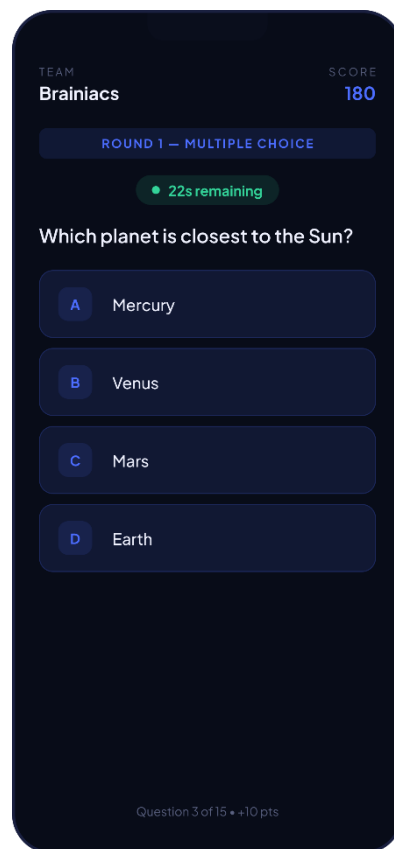
- A QR code will be displayed at the venue (on the venue screen or printed materials).
- The player will scan the QR code, which will redirect them to the game server URL.
- The player will enter the session PIN to join the active game lobby.
- The player will enter their team's name before joining. Solo players will enter their individual name as the team's name.
- Teams can join at any point during the game, including after the session has started, by scanning the persistent QR code visible on every slide.

4.2 Lobby / Waiting Screen

- After joining, the player will see a waiting screen until the host starts the game.
- The player's team name will be displayed at the top of the screen.
- The player will see a confirmation that they are connected and waiting for the host to begin.

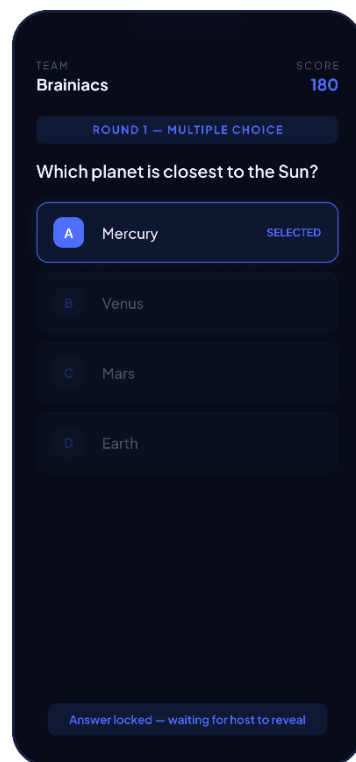
4.3 Question & Answer Screen

- The player will see the question text displayed at the top.
- The player will see 2 to 6 answer options displayed as tappable cards (dynamic per question, not fixed at 4).
- The player will see four answer options displayed as tappable cards.
- The player will tap their chosen answer to submit the response.
- First answer is final — once an answer is tapped, it is locked in and cannot be changed.
- After submission, the interface will show a “waiting for reveal” state, indicating the answer has been recorded.
- Only one submission per question will be accepted per team.
- The countdown timer will be visible on the player's phone (not just on the venue screen).
- For questions with video content, the video will be displayed on the venue screen only. Players will see the question text and answer options on their phones. Video will not play on player devices



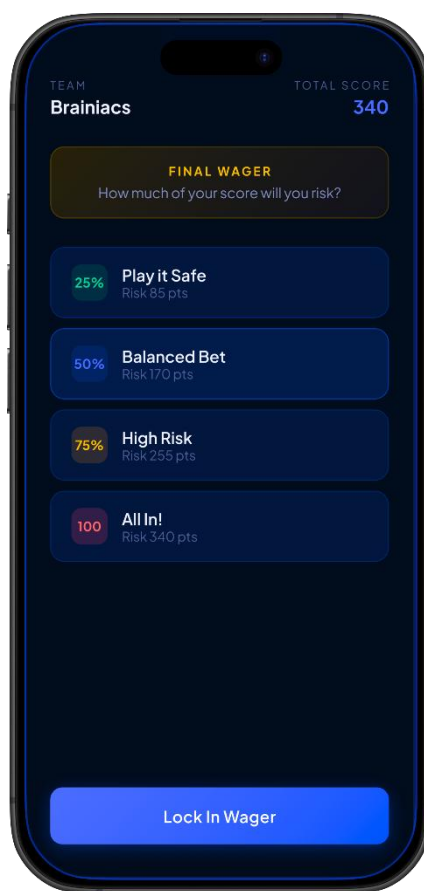
4.4 Answer Reveal & Feedback

- After the host reveals the answer, the player will see whether their response was correct or incorrect.
- Visual feedback will be displayed (e.g., green highlight for correct, red for incorrect).
- The player will see the points gained or lost for that question.



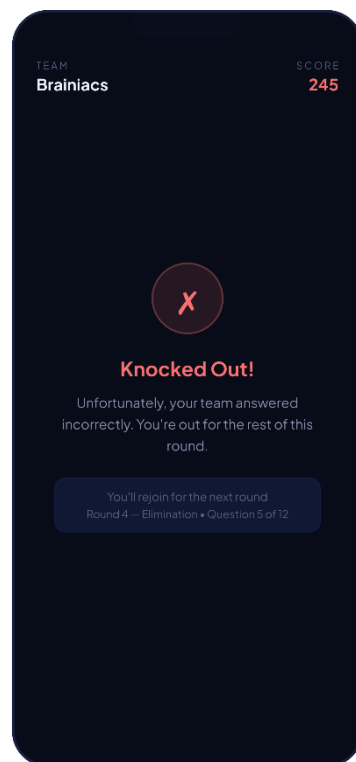
4.5 Wager Input (Wager Rounds)

- Standard Wager: The player will be presented with a wager input where they select a fixed score between 0 and 50 points before the question is displayed.
- Final Wager: The player will select a percentage of their total score (0–100%) to wager on the final question of the night.
- After confirming the wager, the question will be displayed, and the player will answer as normal.



4.6 Elimination Round Behaviour

- If a player/team answers incorrectly during an Elimination round, the platform will automatically disable further participation for that team for the remainder of that round.
- A pop-up message will appear on the eliminated team's phone: "Unfortunately, you've been knocked out until the end of the round."
- Scoring is incremental: Question 1 = 10 pts, Question 2 = 20 pts, up to Question 12 = 120 pts.
- Critical rule: If ALL remaining active teams answer incorrectly on the same question, no teams are eliminated. All active teams lose the points for that question, but proceed to the next question together. Elimination only occurs when some teams answer correctly, and others do not.
- The round continues through all 12 questions. The system will track and display the last team(s) standing at the end.
- The player will see a clear visual indication that they have been eliminated from the current round.
- Eliminated teams will rejoin active play when the next round begins.



4.7 Majority Rules Behaviour

- The player will see the same four answer options and submit their vote.
- After the host reveals the result, players who selected the most popular answer will receive +50 points.
- Players who select a minority answer will lose 50 points.

4.8 3D Mini-Game Interaction

- **Card Shuffle Game:** The player will select which card they believe is the target card (e.g., the Queen) after the 3D shuffle animation.
- **3D Horse Race:** The player will select which horse they wish to bet on via their mobile device.

4.9 Reconnection

- If a player or team disconnects during a live session, they will be able to rejoin using the same session PIN and team name.
- Their score and game progress will be preserved upon reconnection.
- Reconnecting teams can scan the persistent QR code visible on every venue slide or navigate to the URL directly.
- If score data is lost during reconnection, the host can manually re-add the team with their previous score via the Host Control Panel.

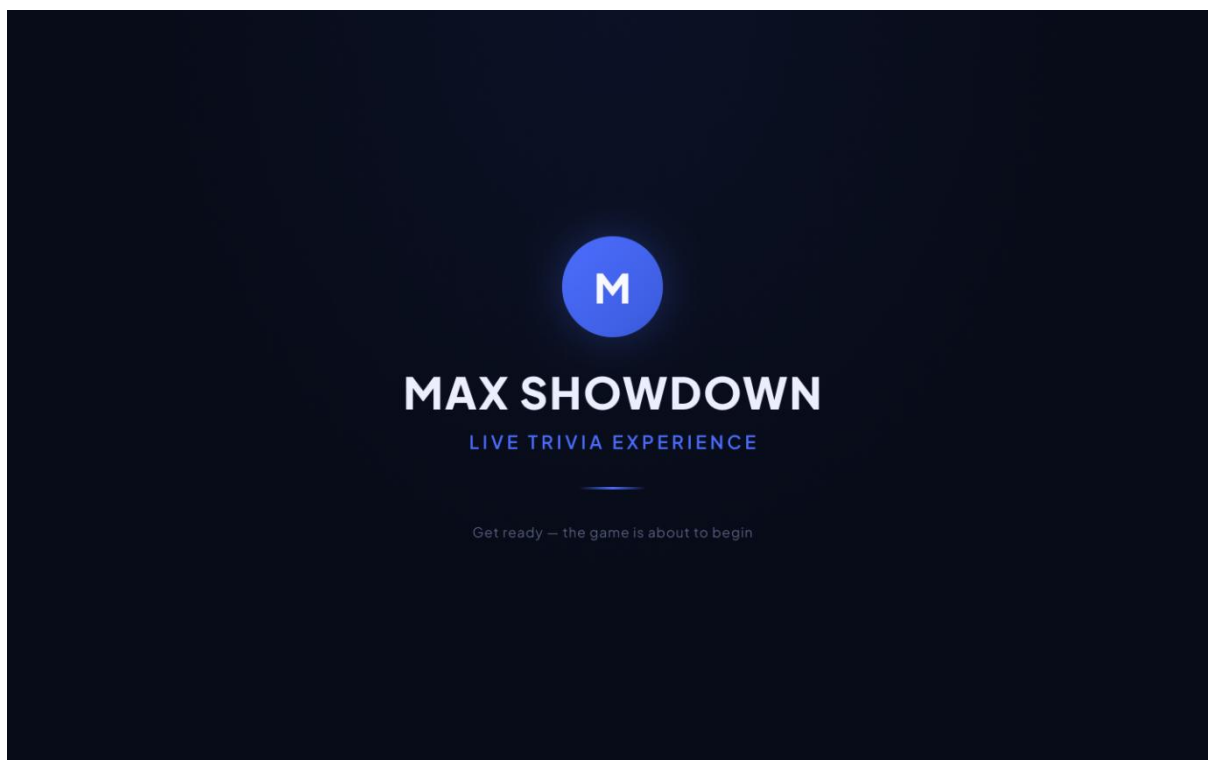
Note: No audio will play on player devices at any point. All sound effects and music playback will be routed exclusively through the venue sound system.

5. Venue Display Screen (Web — Projected/TV)

The Venue Display Screen will be the public-facing visual output of the game, designed for projection on TVs or projectors at live events. This screen is view-only and fully controlled by the host.

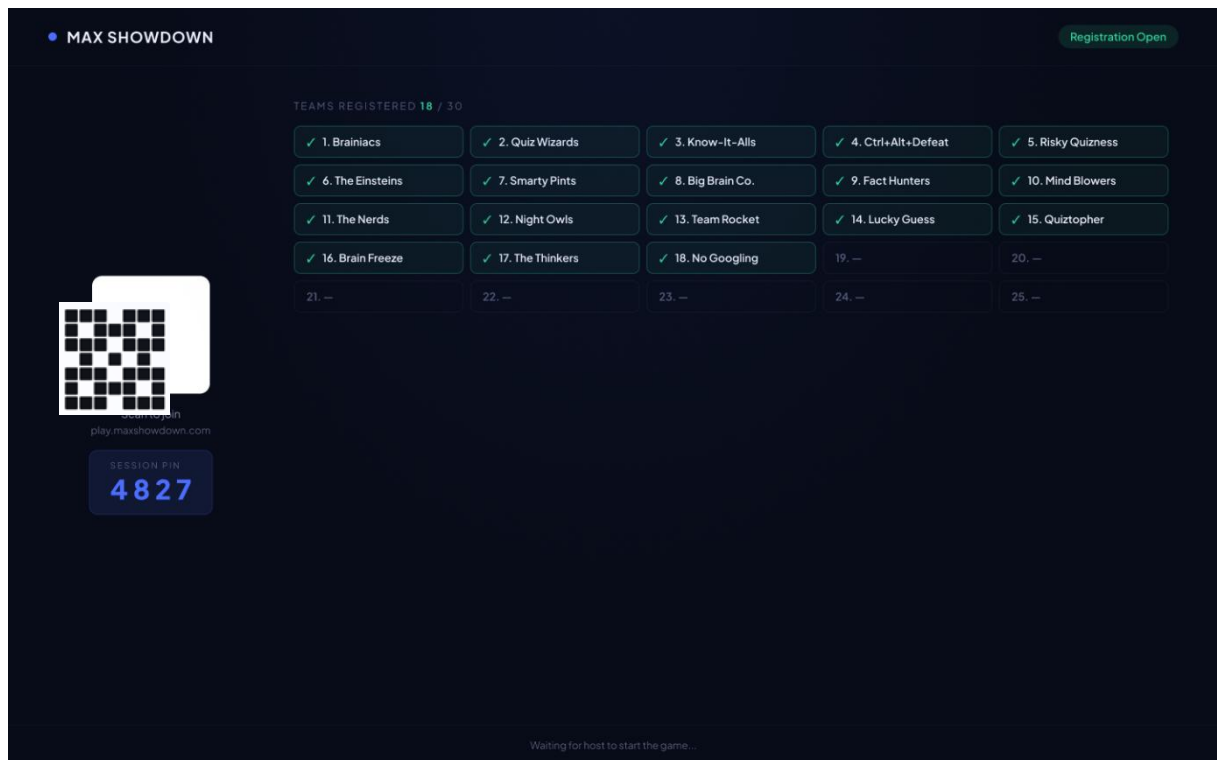
5.1 Welcome Screen (New)

- A branded welcome screen (“Welcome to Max Showdown Live”) with an animated or spinning logo will be displayed while the host sets up and teams register.
- This screen plays in the background before the game starts. It stays until the host advances to the lobby/registration view.



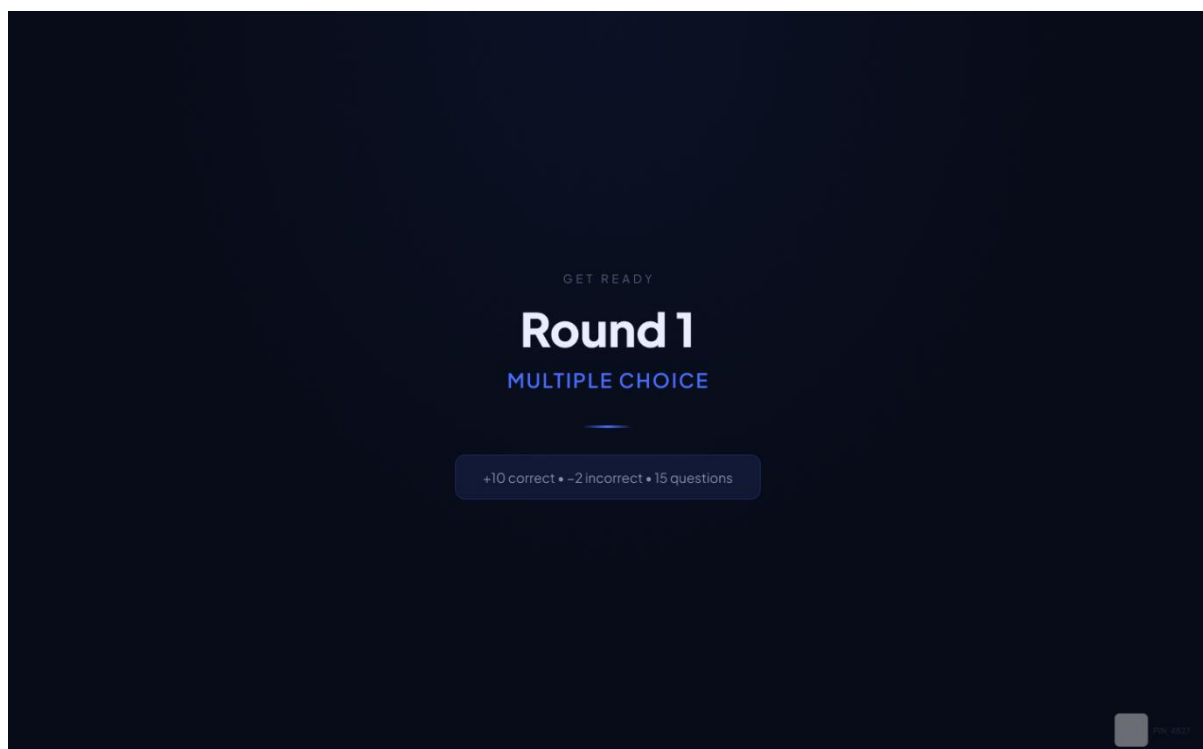
5.2 Team Registration Grid (New)

- Numbered slots (1 to N) will be displayed on the venue screen. As each team joins, their slot turns green with a tick mark.
- Each slot will show the team name alongside the number.
- The host uses this to visually confirm all expected teams are registered before starting the game.
- The host must be able to start the game even if not all slots are filled — late joiners can enter at any time during the session.



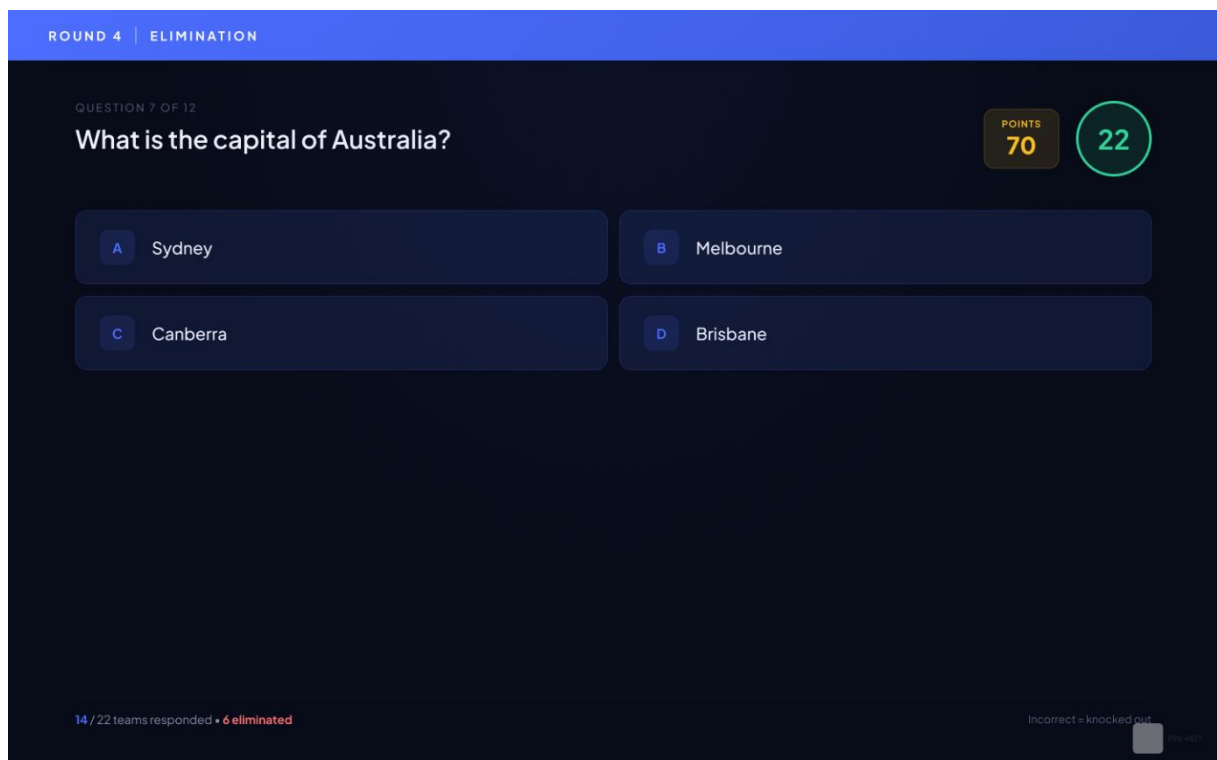
5.3 Round Header

- The venue screen will display a prominent round header banner (e.g., “ROUND 1 | Rapid Fire”) in the electric blue accent bar.
- The round type will be clearly identified so the audience understands the current format.



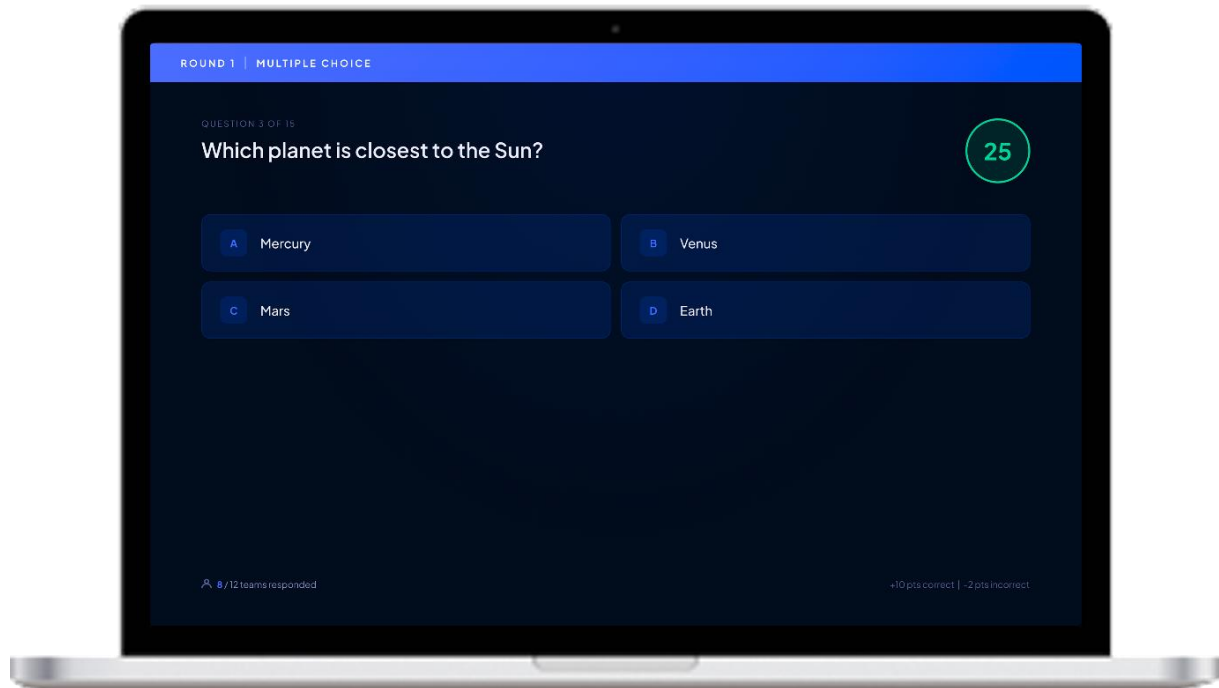
5.4 Round Intro Slide (New)

- Before each round, a dedicated slide will display the round name, round type, and a brief description of the scoring rules (e.g., “+10 correct, –2 incorrect”).
- This slide stays on screen until the host advances. It gives the host time to explain the round to the audience.



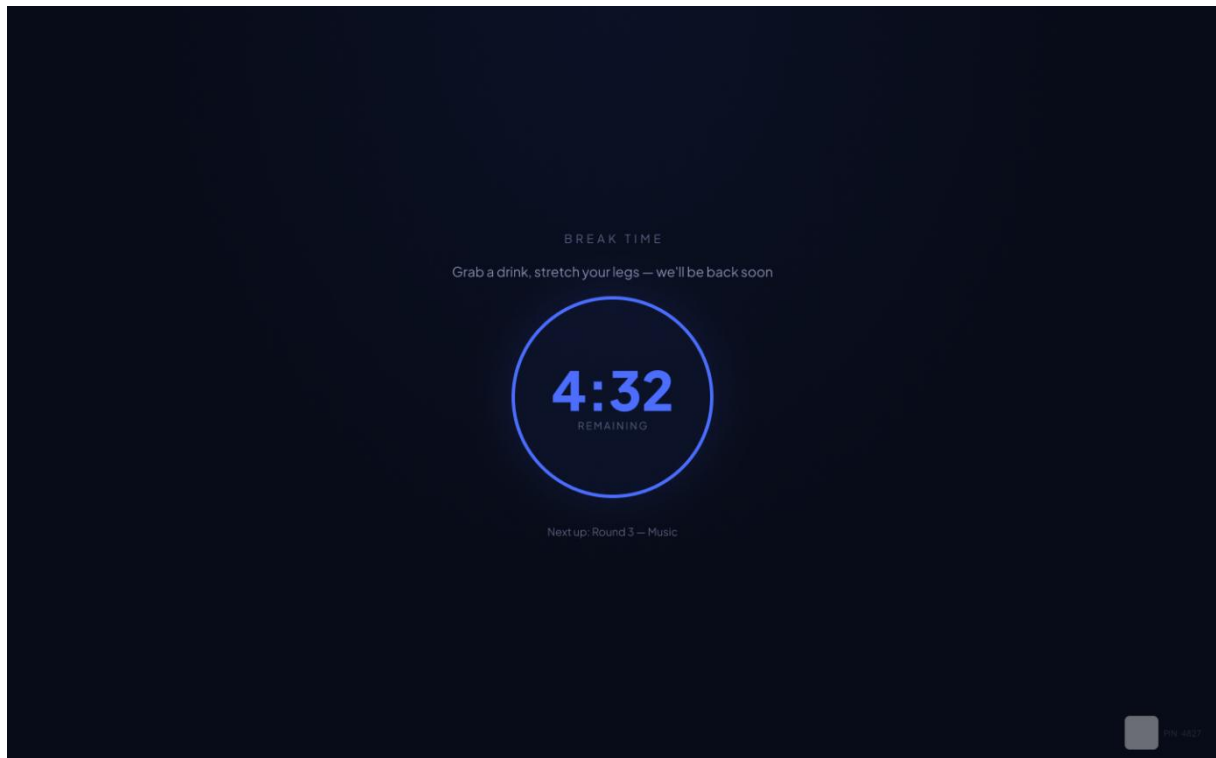
5.5 Question Display

- The venue screen will display the question text in large, readable typography.
- 2 to 6 answer options will be displayed with labels (A, B, C, D, E, F as applicable) in bordered cards.
- The number of points the current question is worth will be displayed on the venue screen.
- If a video clip (MP4) is attached to the question, the video will display on the venue screen alongside or instead of a static image. Video playback is controlled by the host



5.6 Countdown Timer

- A visible countdown timer will be displayed centrally on the venue screen.
- The timer will count down from the configured duration (e.g., 30 seconds as shown in the mockups).
- The timer will be synchronised with the backend to ensure accuracy across all connected interfaces.
- **Timer sound - Timer sound will play through the venue sound system during all rounds except the Music round.**

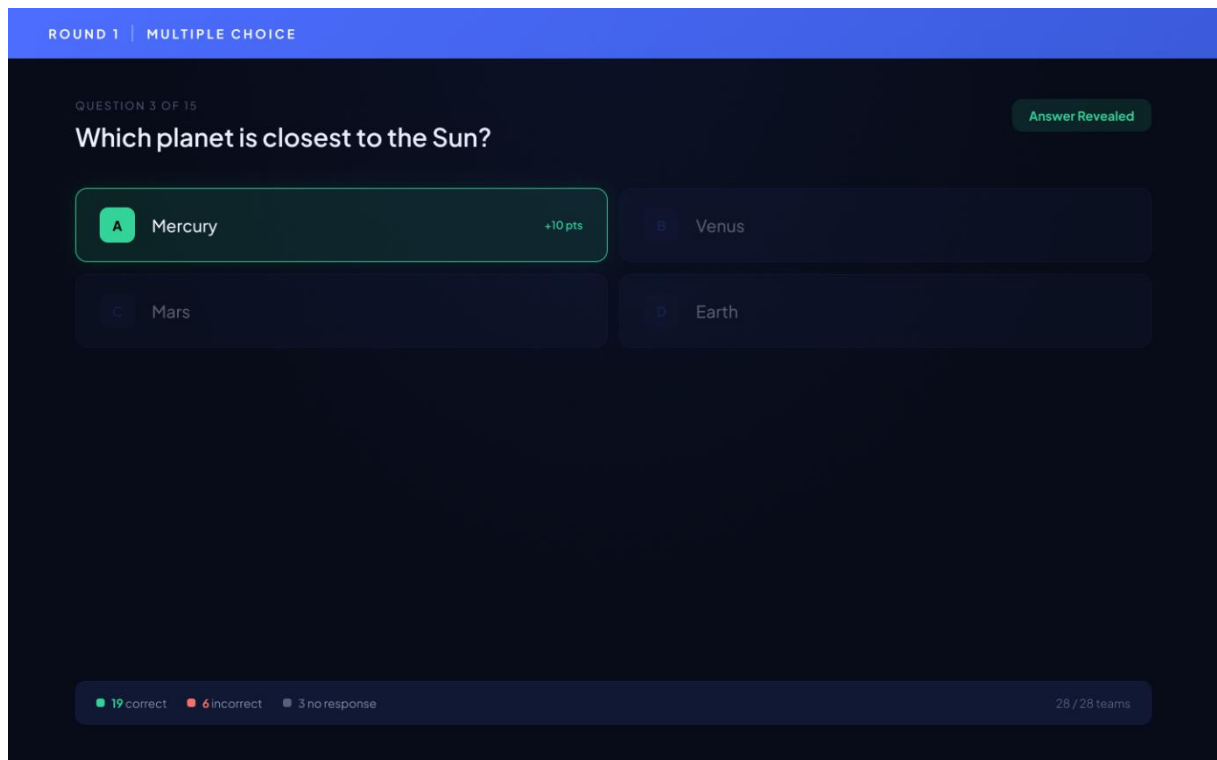


5.7 Live Response Counter

- The venue screen will display a live count of how many teams have submitted their answer for the current question.
- This creates audience engagement and urgency as teams see others submitting before them.

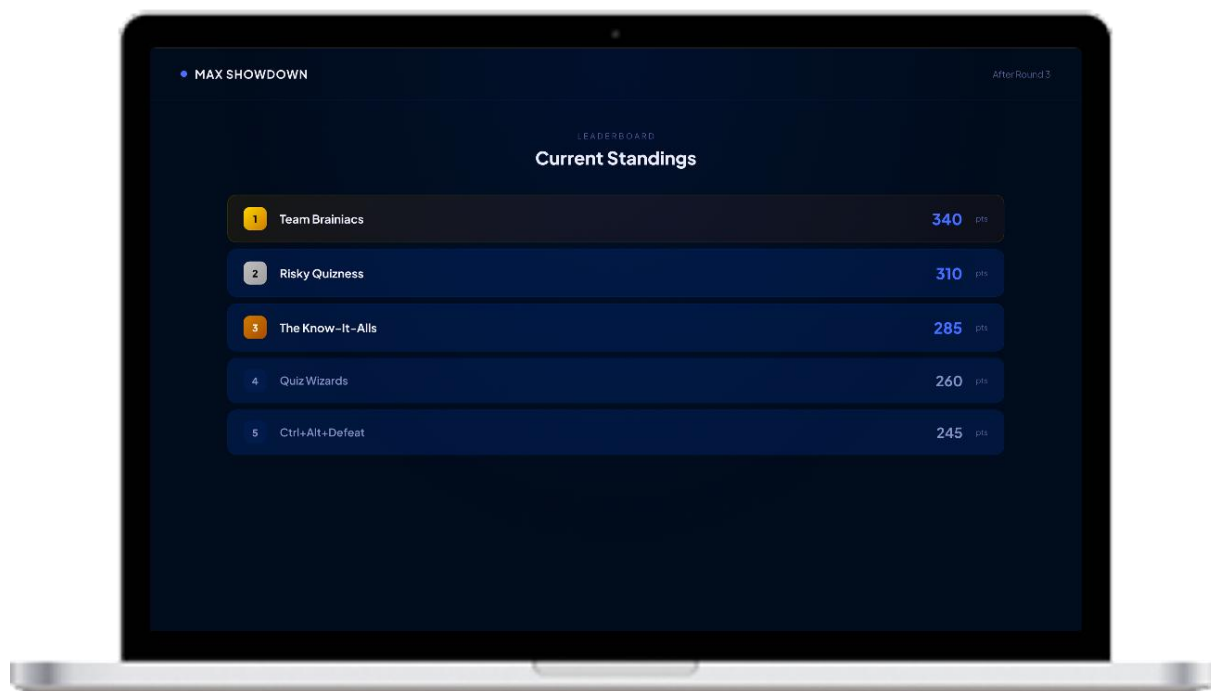
5.8 Answer Reveal

- Upon host trigger, the correct answer will be highlighted on the venue screen.
- Incorrect options will be visually dimmed.
- For Majority Rules rounds, the most popular answer will be highlighted as the winning answer.
- After the reveal, a panel will display how many teams answered correctly, incorrectly, and did not respond.
- Auto-Reveal: If all teams have submitted their answers before the timer expires, the system will automatically reveal the correct answer without waiting.



5.9 Scoreboard Display

- The host will trigger scoreboard display via the S key.
- The venue screen will show the current leaderboard with team names and scores, ranked by position.
- Round-end scoreboards and the final results screen will follow the same display pattern.
- Scoreboards are shown only at the end of each round (not after every question). The host can also trigger an intermediate scoreboard at any time.
- For the final wager question, the host will deliberately NOT show the scoreboard before the last question, to build suspense. The final scoreboard is revealed only after the last answer.



5.10 QR Code Display

- Before the game begins, the venue screen will display the QR code and session PIN for players to join.
- The lobby screen will show teams joining in real time as they scan and connect.
- **Persistent QR Code:** The session QR code will remain visible (e.g., in a corner) on every slide throughout the entire game, allowing late joiners or reconnecting teams to scan at any point.

5.11 Mini-Game Display

- When the host launches a mini-game, the venue screen will display the full mini-game animation (cups shuffle or horse race).
- Results and point awards will be displayed on the venue screen after the mini game concludes.
- **Mini-game scoreboards and results stay on screen until the host presses Next.**

5.12 Break Timer (New)

- **A 6-minute countdown timer will be displayed on the venue screen during breaks.**
- **Clean, elegant design consistent with the dark theme and electric blue accents.**
- **Host-controlled:** the host starts the break timer from the Host Control Panel.
- **Players can see the remaining break time and plan accordingly (e.g., grab food, use facilities).**

6. Round Types & Game Logic

The platform will support seven distinct round types as defined in the Developer Handoff Pack and confirmed by the client. Each round type will have its own game logic module.

The elimination round scoring model has been changed from flat +10/-2 to incremental (10, 20, 30...120). The “all-teams-wrong” rule has been added. Answer options are now dynamic (2–6).

Round	Type	Mechanic	Scoring
1	Multiple Choice	Standard Q&A with four options. One correct answer.	+10 correct, -2 incorrect
2	Wager	Players select a fixed wager (0–50 points) before the question is revealed. Correct = gain wagered amount. Incorrect = lose wagered amount.	Variable: 0 to +50 or 0 to -50
3	Music	Host uploads MP3 files. Audio plays through the venue system. Players answer the related question on their phones.	+10 correct, -2 incorrect
4	Elimination	12 questions with incremental scoring (10, 20, 30...120). Incorrect answer eliminates the team for the round. If ALL remaining teams answer wrong on the same question, no elimination — all lose points but proceed.	Incremental: Q1=10, Q2=20...Q12=120. Eliminated teams score 0 for remaining questions.
5	Majority Rules	The most popular answer among all teams is the winning answer. No pre-set correct answer.	+50 for majority vote, -50 for minority vote
6	Final Multiple Choice	Standard multiple-choice format for the final round.	+10 correct, -2 incorrect
Final	Final Wager	Players wager a percentage (0–100%) of their total accumulated score. Correct = gain that percentage. Incorrect = lose that percentage.	Variable: percentage of total score

7. Scoring Engine

The scoring engine will manage all point calculations across the platform. The base scoring model is fixed, with specific overrides for Wager and Majority Rules rounds.

7.1 Base Scoring

- Correct answer: +10 points
- Incorrect answer: –2 points
- No response (timer expires): 0 points

Note: There is no speed bonus. Points are not affected by response time. There is no fastest-finger mechanic.

7.2 Wager Round Scoring

- Standard Wager: Player selects 0–50 points. The correct answer adds the wagered amount. Incorrect answer deducts the wagered amount.
- Final Wager: Player selects 0–100% of their total score. The correct answer adds that percentage. Incorrect answer deducts that percentage.

7.3 Majority Rules Scoring

- Players who selected the most popular answer: +50 points.
- Players who selected a minority answer: –50 points.

7.4 Elimination Round Scoring (Updated)

- Scoring is incremental per question: Question 1 = 10 pts, Question 2 = 20 pts, continuing up to Question 12 = 120 pts.
- Once a team answers incorrectly, they are eliminated for the remainder of the round and receive 0 points for all subsequent questions.
- Critical rule: If ALL remaining active teams answer incorrectly on the same question, no teams are eliminated. All active teams lose the points for that question but proceed to the next question together.
- The round always continues through all 12 questions. The system tracks and displays the last team(s) standing.

7.5 Leaderboard

- The platform will maintain a running score for each team across all rounds.
- Score updates will be calculated immediately after the host reveals the answer.
- The host will be able to trigger intermediate scoreboards at any point during the game.
- Round-end scoreboards will display team rankings for that round.
- The final results screen will display the complete leaderboard with final standings.

8. Mini-Games

Both mini games have been upgraded from 2D to 3D visuals following the discovery call. The Cups and Balls game has been replaced with a 3D Card Shuffle variant. This upgrade will add approximately 3 additional weeks to the project timeline.

8.1 3D Animal Race (Kangaroo Race or equivalent)

- The venue screen will display an animated 3D race featuring 6 unique kangaroo characters racing on a track.
- Each kangaroo will wear a different colour-coded bib (numbered). The bib colours will match the 6 selection buttons displayed on player phones, so players can easily identify their chosen kangaroo.
- Each kangaroo will hold a bottle of beer (Australian iconic branding) throughout the race.
- Players will select which kangaroo they wish to bet on via their mobile device before the race begins (selecting by colour).
- Pre-race animation: Before the race begins, a starting line animation will show all 6 kangaroos lined up, each taking a sip from their beer bottle. A 3-2-1 countdown will follow, after which the kangaroos begin jumping and running on the track.
- Race duration: 30 seconds.
- Camera angle: Top-down as the primary view. The host will have mouse-based camera control to change angles during the race for added atmosphere.
- Points awarded based on final position. Exact values to be confirmed.
- The scoreboard stays on screen after the race until the host presses Next.
- The race animation and audio will play through the venue screen and sound system only.

8.2 3D Card Shuffle Game (Replacing Cups and Balls)

- 3D card-style visuals — cards shown on a table, flipped face-down, then shuffled.
- Players guess which card (e.g., the Queen) is in the correct position after the shuffle.
- Shuffle speed can increase progressively for added difficulty.
- Players submit their selection via their mobile device.
- After all selections are locked, the correct card is revealed on the venue screen.
- Players who select correctly receive bonus points.

9. Admin Panel & Content Management (Web)

The Admin Panel will be a web-based interface used by the host to create, organise, and manage all quiz content.

9.1 Quiz Builder

- The host will create new quiz sessions by defining rounds, selecting round types, and adding questions to each round.
- For each question, the host will enter the question text, 2 to 6 answer options (dynamic, not fixed at 4), and mark the correct answer.
- The host will configure the round type (Multiple Choice, Wager, Music, Elimination, Majority Rules, or Final Wager) per round.
- The host will be able to upload images (JPG, PNG) and attach them to any question. Images will display on both the venue screen and the player's mobile device alongside the question text and answer options.
- The host will be able to upload MP4 video files (30 Seconds video) and attach them to any question slide. Videos can be used for music video questions, observation-based questions, or any visual content the host wishes to display.
- The host will set the countdown timer duration per question or per round.
- The quiz builder should support a drag-and-drop or dynamic field interface for adding/removing answer options.

9.2 Question Bank

- The host will be able to save questions to a central question bank for reuse across multiple venues and events.
- Questions in the bank will be searchable and categorisable by topic, difficulty, or round type.
- The host will be able to pull questions from the bank into a new quiz session rather than creating them from scratch each time.

9.3 MP3 Upload & Management

- The host will upload MP3 files for use in Music rounds.
- Uploaded audio files will be stored securely and associated with specific questions.
- During the live show, the host will trigger MP3 playback through the venue sound system via the Host Control Panel.
- In addition to MP3 files, the host will upload MP4 video clips via the Admin Panel. Videos will be associated with specific question slides and played on the venue screen during the live show.

9.4 Session Configuration

- The host will generate a unique session PIN and QR code for each game session.
- The host will be able to set the maximum number of teams per session.
- The host will be able to preview the quiz flow before going live.

Note: The host has confirmed that the round structure will remain consistent week to week. Only the question content will be updated regularly. This means the Admin Panel is designed for efficient weekly content refresh rather than structural redesign.

10. Session & Lobby Management

10.1 Session Creation

- The host will create a new game session from the Admin Panel or Host Control Panel.
- A unique session PIN will be generated automatically.
- A QR code will be generated that encodes the server URL with the session PIN.

10.2 Player Join Flow

- Players will scan the QR code displayed at the venue.
- The QR code will redirect to the game URL with the session PIN pre-filled.
- The player will enter their team's name and join the lobby.

10.3 Lobby

- The lobby will display all connected teams in real time as they join.
- The venue screen will show the lobby with the QR code and a live list of connected teams.
- The venue screen will display a team registration grid with numbered slots that turn green as teams join.
- The host will trigger the game to start from the lobby once all teams are connected.
- The host can start the game even if not all expected teams have joined — late joiners can enter at any point.

10.4 Reconnection

- If a player or team disconnects during an active session, they will be able to rejoin by scanning the QR code again or navigating to the URL and entering the same session PIN and team name.
- Their score and round position will be preserved upon reconnection.
- The backend will maintain session state in Redis to support fast reconnection without data loss.
- The persistent QR code on every venue slide enables reconnection at any point during the game.
- If score data is lost, the host can manually re-add the team and set their score via the Host Control Panel.

11. Backend Game Engine

The backend will serve as the authoritative source of truth for all game state, scoring, and timing. It will manage real-time communication between all interfaces via WebSockets.

- **WebSocket Server:** Will manage bidirectional real-time communication between the host, players, and the venue screen using Socket.io.
- **Game State Machine:** Will manage transitions through the defined game flow: Lobby → Round Intro → Question → Timer → Answer Reveal → Score Update → Next Question → Round End → Scoreboard → Next Round → Final Wager → Final Results.
- **Timer Engine:** Server-authoritative countdown synchronised across all connected screens. The host controls start/pause; the server enforces the countdown.
- **Score Calculator (Updated):** Per-round scoring logic including incremental elimination scoring (10–120), wager adjustment, majority rules calculation, elimination tracking, and the “all-teams-wrong” exception rule.
- **Session Manager:** Handles session creation, PIN generation, QR code generation, player join/leave, reconnection, and full session lifecycle.
- **Media Manager:** Serves MP3 and MP4 files to the host/venue interface for playback.
- **Performance Requirements**
 - Minimum 300 concurrent players per session.
 - Session state persisted in Redis for fast access and reconnection support.
 - PostgreSQL for persistent storage of quizzes, question banks, scores, and session history.

12. UI/UX & Design System

The visual design will follow the design philosophy stated in the Developer Handoff Pack: “Apple-like simplicity. Dark theme, electric blue highlights, smooth animations, large typography.”

- Dark theme background (#0B0F1A or similar) across all interfaces.
- Electric blue accent colour for interactive elements, borders, highlights, and round headers.
- Large, clean sans-serif typography optimised for readability at venue projection distances.
- Smooth transitions and animations between game states (question reveal, timer countdown, score updates, answer feedback).
- Responsive mobile layout for the Player Interface, optimised for portrait-mode smartphone usage.
- Widescreen 16:9 layout for the Venue Display Screen, optimised for TV and projector output.
- Minimal, uncluttered interface design — no unnecessary elements during live gameplay.
- Consistent visual language across host, player, and venue screens.
- [3 YouTube reference videos shared by Max during the call will be reviewed by the design team before beginning the UI/UX phase.](#)

- Design reference shared by client shows clear timer, question text, image/photo support, and answer options on the player mobile interface. This layout will be used as a design benchmark.

13. Audio & Sound System

All audio output will be routed exclusively through the host's laptop/venue sound system. No audio will play on player devices.

- Music Round: The host will upload MP3 files via the Admin Panel. During the live show, the host will trigger play/pause from the Host Control Panel.
- Sound Effects: Timer sounds, answer reveal sounds, round transition sounds, and scoreboard reveal sounds will play through the venue system.
- Timer Sound: Audible countdown sound plays through the venue system during all question rounds. Timer sound is automatically muted during Music rounds to prevent interference with MP3 playback.
- Mini-Game Audio: Kangaroos race and cups-and-balls animations will include sound effects routed through the venue system.
- Player Devices: Will remain silent throughout the entire session. No audio files will be pushed to or played on player phones.

Cost Estimate

Estimated Delivery Timeframe	9 Weeks
Option A: Tech Team with AU-based Project Manager	AU \$23,123 + GST
Option B: Tech Team with India-based Project Manager	AU \$19,123 + GST

Payment Schedule

Option A: Tech team with AU-based Project Manager

	Weekly Invoices	Amount
1	From Week 1 to 9	AU \$2,569 + GST
	Total Amount	AU \$23,123 + GST

Option B: Tech team with India-based Project Manager

	Weekly Invoices	Amount
1	From Week 1 to 9	AU \$2,125 + GST
	Total Amount	AU \$19,123 + GST

Estimate Notes

The exact speed of design & development will be affected by client turnaround times, i.e. responding to queries and providing feedback. All services are supplied according to our T&Cs <http://www.dotsquares.com/terms>

The above estimate is based only on the features which we have listed in this document. The estimate is likely to change if you need any other feature (not listed in this doc). It may also vary after sharing your response to any queries being asked in this document.

Note: Your project will be managed & developed by your Dotsquares Team. Other than Developers indicated in a full-time capacity, all team members will be involved partially as required for specific hours in a week.

Resource Utilization

S.no.	Profile	Nature of engagement
1.	One Full Stack Developer	Full-Time
2.	One Game Developer	Partial
3.	One UI/UX Designer	Partial
4.	One 2D/3D Designer	Partial
5.	One Business Analyst/ Project Manager	Partial
6.	One Quality Analyst	Partial

Technology Specifications

Component	Technology
Platform	Browser
Frontend Development	React / Next.js
Backend Development	Node.js
Game Engine	Unity
Design Pattern	Adobe Photoshop / Figma / HTML
Programming Language	JavaScript, C#

We have provided our technology suggestions based on the requirements provided and our experience with the best way of realising these. If you have any specific technical requirements or would like us to explore other options, please just get in touch to let us know.

Client Receivables

S.no	Receivables	Estimated Delivery
1.	Hosting details/SSL Certificate	The date will be mentioned when the project will be awarded.
2.	Domain details will be provided by the client	The date will be mentioned when the project will be awarded.
3.	The client will provide branding assets (If any), any official logo, specific fonts, or colour palettes to be used in the user interface.	The date will be mentioned when the project will be awarded.
4.	Quiz content and MP3 audio files for Music rounds will be provided by the client	The date will be mentioned when the project will be awarded.

Deliverables

S.no	Deliverable	Estimated Delivery
1.	Delivery schedule by BA / PM*(link)	The date will be mentioned when the project will be awarded.
2.	Test Cases by QA* (link)	The date will be mentioned when the project will be awarded.
3.	Project Plan by BA / PM* (link)	The date will be mentioned when the project will be awarded.
4.	Source Code and Web App	The date will be mentioned when the project will be awarded.

Important Points

1. Time estimation may vary if any further functionality is required by you, other than what is mentioned in the above document. Responding to the query may lead to changes in the estimation.
2. We will provide you with two iterations for every UI/UX design. After that, if you want any more iterations, we will consider it as a new requirement, and time may vary.
3. The proposed UI/UX displays in this document are simply mock-ups created using AI Tools; the actual screens may differ.
4. The app will be developed in English for now.

Assumptions

- The platform will be developed as a web-based application using React/Next.js (frontend), Node.js (backend), WebSockets via Socket.io (real-time communication). No native desktop or mobile application will be built.
- All four interfaces (Host Control Panel, Player Mobile Interface, Venue Display Screen, and Admin Panel) will be accessed via a web browser. The Player Mobile Interface will be a responsive mobile web application; no App Store or Play Store submission is required.
- The platform will support team-based play only. Solo players will join as a team of one.
- The two mini-games (Kangaroos and Card Shuffle) will be delivered as 3D web-based animated modules. This is an upgrade from the original 2D scope, confirmed by the client during the discovery call. The 3D upgrade adds approximately 3 additional weeks to the project timeline.
- The Cups and Balls mini game from V1.0 has been replaced with a 3D Card Shuffle variant, as confirmed during the discovery call.
- All audio (sound effects, music round playback, mini-game audio) will play exclusively through the host's laptop/venue sound system. No audio will be pushed to or played on player devices.
- Questions will support 2 to 6 answer options (dynamic per question, not fixed at 4). The quiz builder in the Admin Panel will support this flexibility. This was confirmed by the client during the call.
- First answer is final — once a team submits an answer, it is locked in and cannot be changed. There is no speed bonus; points are not affected by response time.
- The Admin Panel will support quiz creation, question bank management, MP3 upload, and session configuration. Advanced CMS features such as user role management, multi-admin access, or analytics dashboards are excluded from this scope.
- The host will have the ability to manually add a team (with a custom score), remove a team, and edit any team's score during a live session. This was confirmed as a required feature during the call.
- Quiz content (questions, answer options, round structures) will be created and managed by the client via the Admin Panel. Dotsquares will not provide quiz content.
- The platform will be designed to support a minimum of 300 concurrent players per session. The client confirmed typical venue sizes of 200–300 people with approximately 25–30 teams per event.

- The elimination round will consist of 12 questions with incremental scoring (10, 20, 30... up to 120). If all remaining active teams answer incorrectly on the same question, no teams are eliminated — all lose points but proceed. This specific rule was confirmed during the call.
- No offline mode is required. The platform will require an active internet connection for all interfaces (host, player, and venue screen).
- Post-event features such as result exports, historical leaderboards, or event analytics are excluded from this scope unless confirmed otherwise during development.
- White-labelling and multi-host licensing capabilities are excluded from this scope. A potential Phase 2 SaaS/licensing portal (Super Admin dashboard with multi-business onboarding, subscription payments, and usage monitoring) was discussed during the call and will be scoped separately after MVP delivery.
- Browser compatibility will target the latest versions of Chrome, Safari, Firefox, and Edge. Legacy browser support is excluded.
- The QR code and session PIN mechanism assumes all players have access to a smartphone with a modern mobile browser and an active internet connection at the venue.
- A persistent QR code will be displayed on every venue slide throughout the game, allowing late joiners or reconnecting teams to scan and join at any point. Teams can join after the session has started.
- The venue screen will include additional features: a welcome screen, team registration grid, round intro slides, a 6-minute break timer, live response panel, auto-reveal when all teams respond, and points-per-question display. These were confirmed during the call.
- The host confirmed that branding assets (logo, fonts, colour palette) will be provided before the design phase begins. 3 YouTube reference videos were shared during the call for the design team to review.
- When the host presses Next/Space, the timer starts, answer options appear on player phones, and (for music rounds) the MP3 begins playing — all simultaneously. No slide, question, or scoreboard will auto-advance without explicit host action.
- The platform will support MP4 video file upload and playback on question slides. Video will play on the venue screen only, not on player devices. No third-party video streaming (YouTube, Vimeo) integration is included — MP4 files only.

Change Request Process

We understand that during the process of Software Development, the client may need to add or alter requirements due to changing market needs. To ensure that the clients can get the best products developed, we at Dotsquares have an established CR process that integrates these changes into ongoing development seamlessly.

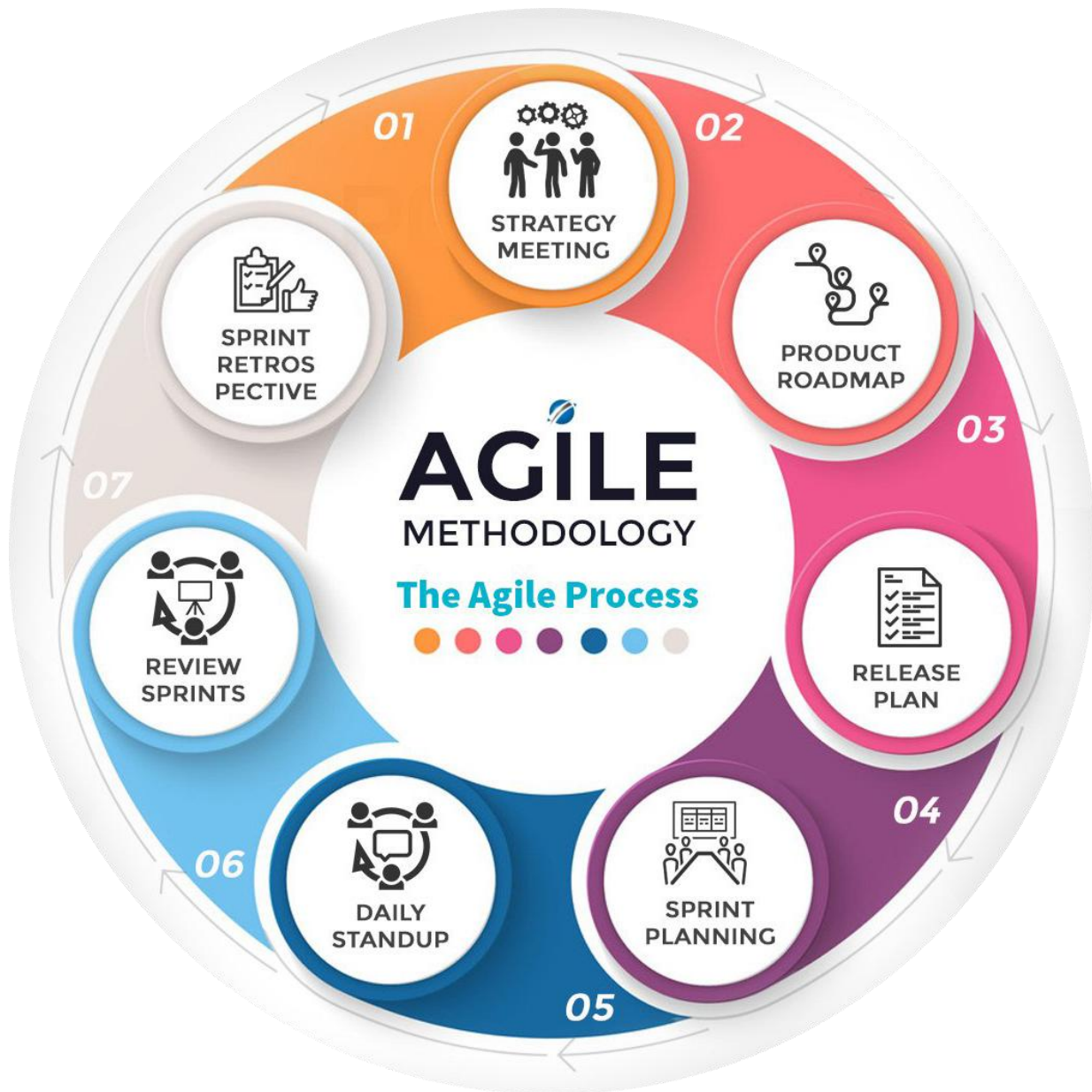
What is a CR?

A change request is a proposal to alter a product or system, often brought up by the client. During a project, this can happen when a client wants to change or alter the agreed-upon deliverables. Thus, any requirement or work not mentioned under the Scope of Work section in this document will constitute a change request.

Our CR process follows a set of predefined steps as shown below



Software Development Life Cycle:



Why is SDLC time mandatory?

Dotsquares is a CMMi L3 Process-based organization, so every process has been defined in the organization, and our project managers always try to control all project constraints.

As per the diagram, every project has a few constraints to control the project, such as scope, time, cost, quality, risks, and resources, if any constraints go beyond the limit or failed that project may get

failed. The project manager and their supporting team control all the constraints from initiation to closing of the project to make sure that we deliver the best quality product and services to our client.

Note: - Project manager, technical lead, and Quality analyst are involved to control the project constraints through

- **Project Manager** - Project Planning, SRS Document, Sprint Planning, Team Meetings, Communication, UAT, etc.
- **Technical Lead** - Database Schema Designs, System Design Documents, Code Review, Unit Testing, Technical Assistance to Development Team, Team Meetings, Technical Communication, UAT, etc.
- **Quality Analyst** - Functional Testing, Create Test Cases, Test Cases Execution, System Testing, Regression Testing, Non-Functional Testing, Browser Compatibility, Responsive Testing

This will be our project management process in steps:

- **Kick-off Meeting:** As soon as the project is on board, we take a kick-off meeting on the very first day with all the team members' involved Project manager, developer (s), Designer, and Client. The agenda of this meeting is to make sure all the stakeholders involved are on the same ground in terms of requirements, discuss the details required like payment gateway details and API credentials, and discuss the time for the regular update meetings.
- **SRS:** This is the System Requirement Specification document that entails the complete requirement in utmost detail and the acceptance criteria. This is a more polished version of the scope document that we share at the time of estimation. SRS is complete in approx. 2 days.
- **Design:** Design and SRS start from day one simultaneously. We will provide you with 2 iterations for every page design, after that, if you want any more iterations, then we will consider it as a new requirement, and time may vary.
- **Development:** Development also starts from day one, but while the PM is working on SRS, the development team works on setting up the development environment, staging/testing server so the client can view the regular updates, Database design, etc., and once the SRS completes (SRS completes in approx. 2 days) the team works on the project as specified in the SRS.
- **Client engagement and communication:** Dotsquares has its system to maintain proper communication with clients which is DSP (Dotsquares Projects). Any of the team members can create different threads in DSP, and other people involved can respond. Each message

posted on DSP sends an e-mail to all the people involved to keep them up to date. You can also connect with the project manager on Skype or call in case of any queries or doubts.

- **Updates:** The team will share daily updates about the work they have done on the day and post the message on DSP. As soon as the concerned module is completed, the team will put it on the staging server so the client can review the work done and share their feedback. The team will work on the feedback accordingly. There will be multiple releases during the project development, so the client is aware of the system.

Additional Services

Support & Maintenance

At an additional cost, we offer support and maintenance under the following options:

Bucket

- It is a good option when ad-hoc support is required.
- The client can purchase 20/40/60 hours on a pre-pay basis.
- The client can utilise the purchased hours when required.
- There will be no monthly fees or recurring contacts.
- Only the time used will be removed from the bucket, in amounts as small as 15 Minutes.
- The bucket is kept with the developer/PM & authorisation is not required from our accounts.
- Team & Accounts will keep you posted on Bucket hours, and the same will be shared on our Project Management System.
- You can top up the bucket with the same or fewer hours when purchased hours are exhausted.

Retainer

- It's a good option when a certain number of hours of support is required monthly, and our 24X7 support is required.
- No contracts. The engagement can be terminated at any time.
- The day-to-day activity log of the allocated members will be shared with you via our Project Management System.

Warranty

- If any bug occurs or any feature is not working in the game, we will resolve that issue within 30 days of completing the project. If you want any additional features, then we will estimate you accordingly.

We can also design a custom support maintenance plan to suit your business and project requirements. To find out which plan is more suited for your project, please contact your Project Manager.

Development Process

Project Management Tool

Dotsquares will provide free access to APMS, our project management and collaboration platform for project planning, communication, and tracking.

- Daily updates provided
- Keeps digital track of conversations
- Milestones and tasks for planning
- Maintain task list for deliverables
- E-mail notifications about updates & other activities



Testing

We provide unit testing services for your project as a standard package. Unit testing is completed by the developer during the development cycle. We ensure all the software code blocks (paths) are executed at least once and produce the desired results.

Environment

Where possible, Dotsquares provides free hosting during the project for you to test at your convenience. If the project dictates specific environmental requirements, a separate estimate will be provided for these.

When the project is finished and signed off, we ask you to choose a hosting provider and deploy it to the live environment.

Deployment

Dotsquares will deploy the complete solution for the project. Code can also be supplied directly to you or any repository account held.

Web-based developments will normally be transferred to your hosting account and put live as required. Mobile applications will be uploaded to the relevant App Store under your developer account.

Warranty

Your software is delivered with a free warranty of 30 days. You must notify the team during the 30-day warranty period if you identify an error in the code/system/program. Once it has been agreed upon, Dotsquares will correct any errors made in the construction of the code/system program.

Intellectual Property Rights

All the Intellectual Property created, written, developed, furnished, or produced by us during the term of the agreement/project, within the scope of the agreement or any applicable Statement of Work, shall be the client's exclusive property.

CMMI

What our CMMI Maturity level rating means for you.

Improved Quality

- Well-defined processes limit opportunities for error.
- Smart supplier management ensures quality results.
- Strategic people management enables team members to develop skills that will help the Organisation succeeds.

Decreased Costs and Enhanced Productivity

- Streamlined processes allow teams to integrate and collaborate more efficiently.
- Standardised project management practices reduce redundancy and waste.
- Identifying skill gaps helps break down workflow bottlenecks.

Higher User Satisfaction

- Service delivery strategies strengthen weak User touch points.
- Elective product development instills User confidence.
- Project management practices help teams meet and manage User expectations.

Improved On-Time Delivery

- Field-tested project management standards keep work on schedule.
- Clear development processes reduce scope creep and deadline extensions.
- Careful vendor management mitigates supplier delays.

Why Dotsquares?

Dotsquares are an international creative digital design and development company striving to make your vision a reality.

We thrive on seeing ideas grow. Since 2002, we have been working with individuals and global brands to embrace the evolving digital industry.

We are big on trust, reliability and mutual respect so we will take the time to understand you, your product and your business. We work closely with all our clients and are always looking to forge ongoing trusted relationships.

What we do...

WEB



Our web development professionals bring years of experience to meet both the business and technical requirements of a project.

GAMES



Our game developers use platforms such as Unity, Maya, Cocos2D, Unreal, AR Kit, Maya, Max, and more to create the best gaming experience.

APPS



We provide cutting-edge mobile app development that can efficiently and creatively meet current business demands.

CRM



Each company is different, and we can create bespoke CRM systems to manage our client relationships

DIGITAL MARKETING



The industry is becoming saturated. Using methods like SEO, SMM, PPC, and ads, our marketing experts can give your brand a voice.

IT SUPPORT



Server support, network troubleshooting, remote installation, and our hosting experts are on hand 24/7 to keep your business alive.

CLIENTS

Trusted by Global Leaders

nikpol
red rooster

HONDA

shieldpay
JUST EAT

airbnb
MONEY

DUNKIN' DONUTS

THOMSON REUTERS
Allurion
FUJITSU

AZETS
FISHER & PAYKEL


OUR OFFICES

Global Offices with Onshore Teams

Dotsquares operates globally with offices in


Australia

(+61) 3 9511 5598


UK, London

(+44) 208 0901819


UK, Brighton

(+44) 1273 575190


USA

(+1) 301 563 9488


India

(+91) 98290 29555

What our Clients says

Daniel Ox

Ocean Holidays Group
UK

"Significant cost savings can be made by working with an outsourced development company you can trust. Dotsquares have an exceptional team of quality engineers who work quickly and are always aware of deadlines."

John Wussah

Education & Training Company
Australia

"You have been one of the best developers I have ever worked with. You have excellent technical and interpersonal skills and you got my website development and design project is done to my amazement and satisfaction!"

Brian Riback

New York Daily News
USA

"Dotsquares went beyond the requirements. The team is amazing and working with them is a winning formula. I feel extremely lucky to have these gentlemen on my team with their unyielding dedication to excellence."